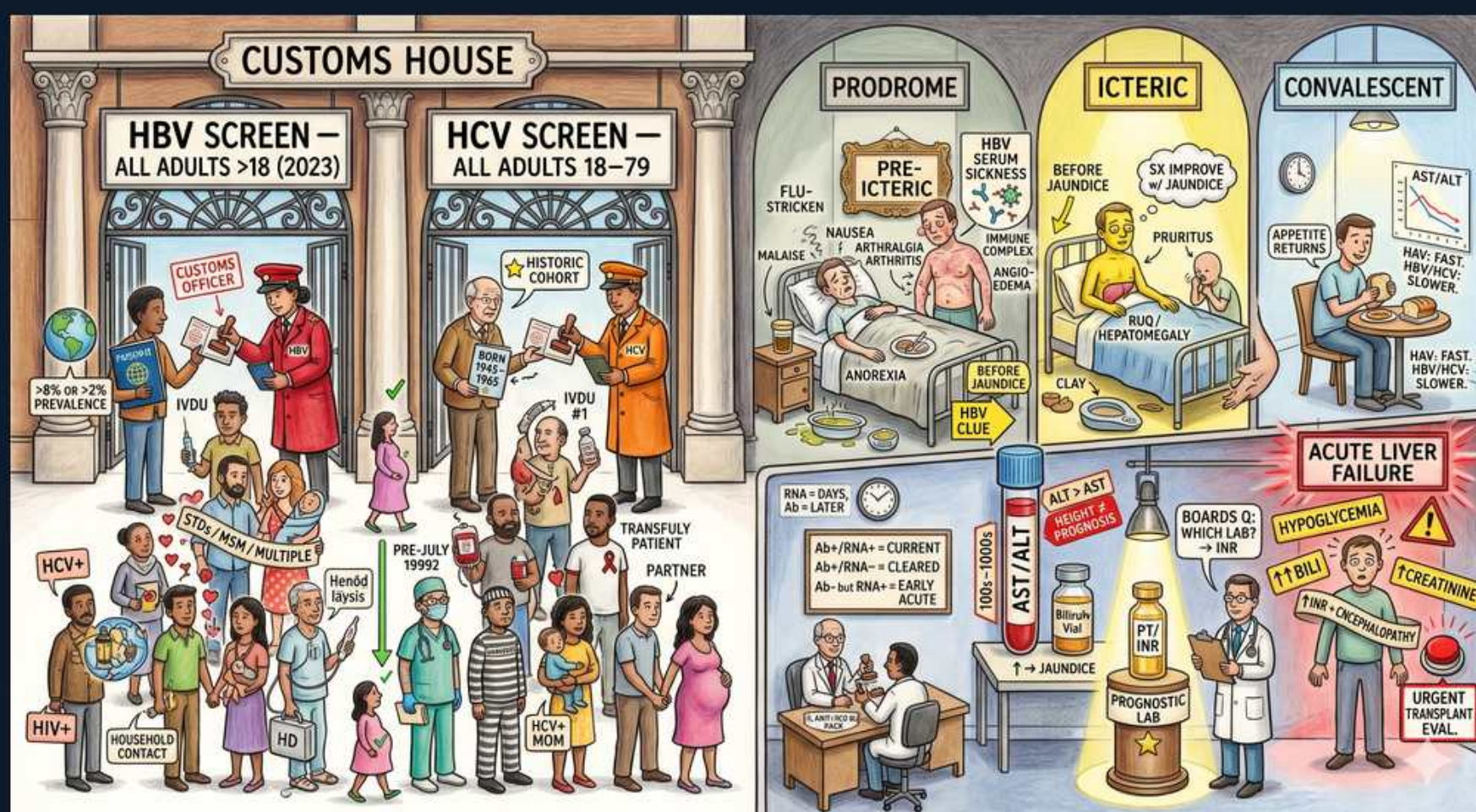


Viral Hepatitis — Hepatitis C & Screening



1. HCV screen — all adults 18–79

WHAT YOU SEE

HCV screen gate, all adults 18–79 (2023)

WHAT IT ENCODES

USPSTF/CDC recommend ONE-TIME HCV screening for ALL adults aged 18-79, plus periodic screening for ongoing-risk groups (IVDU, HIV, dialysis) and screening in EVERY pregnancy. The screening test is anti-HCV antibody (EIA); positives are reflexed to HCV RNA to confirm active infection.

BOARD TRAP

Current guidance is UNIVERSAL screening, not risk-based only — screen even patients who deny risk factors. A stem saying 'no risk factors, so don't test' is the wrong answer.

2. IVDU — leading risk

WHAT YOU SEE

IVDU figure at HCV gate

WHAT IT ENCODES

Injection drug use is the most common route of HCV transmission today. Sharing needles/equipment transmits efficiently because HCV is parenterally spread. Intranasal cocaine use (shared straws) and unregulated tattoos are additional parenteral routes.

BOARD TRAP

HCV is primarily PARENTERAL/blood-borne and far less sexually transmitted than HBV. Don't choose 'sexual contact' as the leading HCV route — that's a better fit for HBV.

3. Pre-1992 transfusion

WHAT YOU SEE

historic cohort, pre-July 1992 transfusion

WHAT IT ENCODES

Blood transfusion before donor anti-HCV screening began (mid-1992) is the classic historical HCV exposure — a key reason for the original birth-cohort (1945-1965 'baby boomer') screening recommendation now superseded by universal screening.

BOARD TRAP

Post-1992 transfusion HCV risk is negligible due to nucleic-acid blood-supply screening. A current transfusion is not a meaningful HCV risk factor — don't attribute new HCV to a recent transfusion.

4. Transfusion / dialysis patient

WHAT YOU SEE

transfusion patient figure

WHAT IT ENCODES

Chronic hemodialysis patients and prior blood-product recipients (e.g., clotting-factor concentrates before viral inactivation) carry elevated HCV prevalence and warrant periodic screening.

BOARD TRAP

Occupational needlestick HCV seroconversion risk is ~1.8% — higher than HIV (~0.3%) but much lower than HBV (up to ~30% from an HBeAg+ source). Don't rank HCV as the highest needlestick risk; HBV is.

5. Low vertical / sexual transmission

WHAT YOU SEE

mother & partner figures, lower transmission

WHAT IT ENCODES

Perinatal HCV transmission is ~5% (rising with high maternal viremia or HIV co-infection), and sexual transmission is uncommon in monogamous heterosexual couples. These contrast with the much higher vertical/sexual efficiency of HBV.

BOARD TRAP

Breastfeeding is NOT contraindicated in HCV (unless nipples are cracked/bleeding). Don't tell an HCV+ mother to avoid breastfeeding. Vertical and sexual risk both rise with HIV co-infection.

6. Anti-HCV → confirm with RNA

WHAT YOU SEE

lab table: Ab+/RNA+ current, Ab+/RNA- cleared

WHAT IT ENCODES

A positive anti-HCV antibody indicates exposure but NOT necessarily active infection; it must be confirmed with HCV RNA (PCR). Ab+ / RNA+ = current active infection; Ab+ / RNA- = past/cleared or successfully treated infection (antibody persists for life).

BOARD TRAP

Ab+ / RNA- = resolved/cured infection, NOT active disease — the antibody stays positive after spontaneous clearance or DAA cure, so a positive antibody alone never proves current infection. Always reflex to RNA.

7. Ab- / RNA+ = acute

WHAT YOU SEE

lab line Ab-/RNA+ acute window

WHAT IT ENCODES

In early acute HCV, HCV RNA becomes detectable within 1-2 weeks of exposure, well BEFORE anti-HCV antibody seroconverts (which can take 8-12 weeks). So an antibody-negative, RNA-positive result indicates acute infection in the window.

BOARD TRAP

A NEGATIVE HCV antibody does not exclude acute HCV — if exposure was recent (e.g., needlestick, new IVDU), order HCV RNA. Relying on antibody alone misses acute window-period infection.

8. High chronicity rate

WHAT YOU SEE

clinical course bays, chronic progression

WHAT IT ENCODES

Approximately 75-85% of acute HCV infections progress to chronicity — far higher than adult-acquired HBV (~5%). This high chronicity, combined with a usually silent acute course, makes HCV the leading cause of chronic viral hepatitis and cirrhosis in the West.

BOARD TRAP

HCV is usually asymptomatic acutely yet MOST cases become chronic — the opposite pattern from adult HBV (often symptomatic acutely but mostly cleared). Don't swap the chronicity rates of HBV and HCV.

9. Often asymptomatic acute

WHAT YOU SEE

prodrome/icteric bays, mild course

WHAT IT ENCODES

Acute HCV is frequently silent — anicteric and minimally symptomatic — which is precisely why universal screening (rather than waiting for symptoms) is essential to catch it before chronic complications develop.

BOARD TRAP

Fulminant acute liver failure from HCV alone is RARE — HCV is a chronic-disease problem. If a stem describes acute liver failure, think acetaminophen, HAV, HBV±HDV, or HEV, not isolated acute HCV.

10. Cirrhosis progression

WHAT YOU SEE

clinical corridor toward fibrosis

WHAT IT ENCODES

Chronic HCV drives slowly progressive hepatic fibrosis to cirrhosis over ~20-30 years; roughly 20-30% of chronically infected patients develop cirrhosis. Fibrosis can be staged noninvasively (FibroScan/elastography, FIB-4) to guide surveillance.

BOARD TRAP

Cofactors — ALCOHOL, HIV/HBV co-infection, and hepatic steatosis (obesity/diabetes) — markedly ACCELERATE fibrosis. Counseling on alcohol cessation is part of HCV management; don't ignore alcohol as a modifier.

11. HCC risk

WHAT YOU SEE

complication panel, liver cancer

WHAT IT ENCODES

HCV-related cirrhosis is a major cause of hepatocellular carcinoma. Patients with cirrhosis need HCC surveillance with abdominal ultrasound (± AFP) every 6 months. Extrahepatic HCV associations to recall: mixed cryoglobulinemia, membranoproliferative glomerulonephritis (MPGN), and porphyria cutanea tarda.

BOARD TRAP

HCC surveillance (US ± AFP every 6 months) CONTINUES even AFTER DAA cure if cirrhosis is already established — achieving SVR lowers but does not eliminate HCC risk. Stopping surveillance after cure in a cirrhotic is the trap.

12. Cure with DAAs

WHAT YOU SEE

HCV treatment algorithm / DAA regimen

WHAT IT ENCODES

Direct-acting antivirals (e.g., sofosbuvir/velpatasvir, glecaprevir/pibrentasvir) cure >95% of HCV across genotypes with 8-12 week oral, well-tolerated courses. Cure = SVR12 (sustained virologic response, undetectable HCV RNA 12 weeks after finishing therapy). Pangenotypic regimens have made routine genotyping often unnecessary.

BOARD TRAP

SVR (cure) does NOT reverse established cirrhosis, and a cured patient can be re-infected — antibody stays positive, so re-infection is diagnosed by RNA, not antibody. HCC and varices surveillance must continue in cirrhotics despite cure.

13. Acute liver failure — danger room

WHAT YOU SEE

red ALF danger room, INR/creatinine/encephalopathy

WHAT IT ENCODES

Acute liver failure = coagulopathy (INR ≥1.5) PLUS hepatic encephalopathy in a patient without pre-existing cirrhosis, evolving over days-weeks. Management is ICU care plus urgent transplant evaluation; give N-acetylcysteine for acetaminophen (and consider in non-acetaminophen) ALF.

BOARD TRAP

ALF is RARE in HCV — when you see fulminant hepatitis, think HAV/HBV/HEV or acetaminophen, not HCV. Also remember encephalopathy is required to diagnose ALF; coagulopathy alone is not enough.

14. HBV screen — all adults ≥18

WHAT YOU SEE

HBV screen gate, all adults >18 (2023)

WHAT IT ENCODES

For contrast: 2023 CDC guidance also recommends screening ALL adults ≥18 at least once for HBV. The proper HBV screen is a TRIPLE panel — HBsAg, anti-HBs, and anti-HBc (total) — to classify infection, immunity, and susceptibility in one shot.

BOARD TRAP

Don't conflate the two screens: HCV is screened with anti-HCV antibody (then RNA), whereas HBV needs the triple panel (HBsAg/anti-HBc/anti-HBs), NOT just a single antibody. Screening HBV with anti-HBs alone misses active infection.